Title of the Notes

Remote Group

For November 15th

Problem (Munkres 55.4)**.** Suppose you are given the fact that for each n, there is no retraction $r : B^{n+1} \to S^n$. Prove the following:

(a) The identity map $i : S^n \to S^n$ is not nullhomotopic.
(b) The inclusion map $j : S^n \to \mathbb{R}^{n+1} - 0$ is not nullhomotopic.
(c) Every nonvanishing vector field on B^{n+1} points directly outward at some point of S^n, and directly inward at some point of S^n.
(d) Every continuous map $f : B^{n+1} \to B^{n+1}$ has a fixed point.
(e) Every $n+1$ by $n+1$ matrix with positive real entries has a positive eigenvalue.
(f) If $h : S^n \to S^n$ is nullhomotopic, then h has a fixed point and h maps some point x to its antipode $-x$.

Proof of (a). We will prove by contrapositive.

Apologies if there are some slips in the rigor here, this was hard.

Suppose the identity $i : S^n \to S^n$ is nullhomotopic.

Then there is a continuous map

$$H : S^n \times I \to S^n$$

such that

$$H(x,0) = x,\ H(x,1) = p$$

for some fixed $p \in S^n$.

Starting with $S^n \times I$ we can impose $(x,1) \sim (y,1)$ for all $x, y \in S^n$. H is then constant on all equivalence classes, letting us factor

$$\begin{array}{ccc} S^n \times I & \xrightarrow{H} & S^n \\ & \searrow^{q} & \uparrow{\scriptstyle \varphi} \\ & & (S^n \times I)/(\sim) \end{array}$$

giving a continuous map φ for which the diagram commutes (q is $q(x,t) = [(x,t)]$).

For this map we have that

$$\varphi \,|_{S^n \times 0} = H \,|_{S^n} = \mathrm{id}_S^n .$$

Now it remains to show that

$$(S^n \times I)/\sim \cong B^{n+1}.$$

Take $f : (S^n \times I)/\sim \longrightarrow B^{n+1}$ by

$$f([(x_1, x_2, \dots, x_{n+1}), t]) = (1-t)(x_1, x_2, \dots, x_{n+1}).$$

Then the point p corresponding to the nontrivial equivalence class is the origin.

The map is a well-defined homeomorphism (I swear!).

This gives a retraction map from $B^{n+1} \to S^n$ which we were told wasn't possible! ■

Proof of (b). I'm sure one could argue this from the fact that $S^n \simeq \mathbb{R}^{n+1} \setminus 0$, but instead, take the retraction

$$r : \mathbb{R}^{n+1} \setminus 0 \to S^n,\ r(x) = \frac{x}{\|x\|}.$$

Observe that

$$r \circ j = \mathrm{id}_{S^n}.$$

Suppose, for contradiction, that j is nullhomotopic. Then

$$j \simeq c_p$$

for some constant map $c_p : S^n \to \mathbb{R}^{n+1} \setminus 0$.

Composing with r, we get

$$\mathrm{id}_{S^n} = r \circ j \simeq r \circ c_p,$$

where $r \circ c_p$ is constant, contradicting (a). ■

Problem. A nonvanishing vector field on B^{n+1} is a continuous map

$$v : B^{n+1} \to \mathbb{R}^{n+1}$$

such that

$$v(x) \neq 0\ \forall x \in B^{n+1}.$$